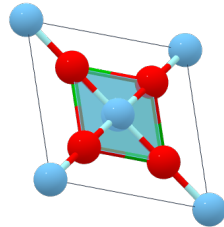
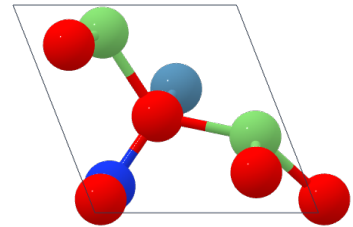


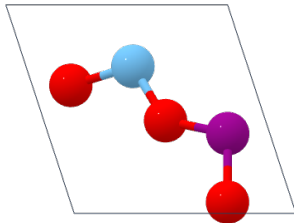
$\text{BaV}_2\text{Ni}_2\text{O}_8$
 $\text{BaTi}_2\text{V}_2\text{O}_8$
 -3.09 eV/atom



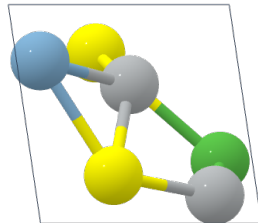
SrTiO_3
 BaTiO_3
 -3.56 eV/atom



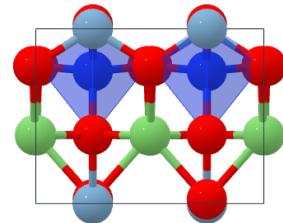
$\text{Li}_2\text{CaSiO}_4$
 $\text{Li}_2\text{CaSiO}_4$
 -3.03 eV/atom



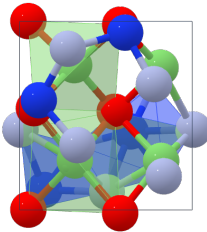
Co_2TiO_4
 TiMnO_3
 -2.38 eV/atom



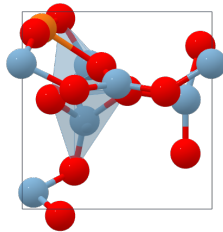
ErNi_2Ge_2
 $\text{LaAl}(\text{NiS})_2$
 -1.19 eV/atom



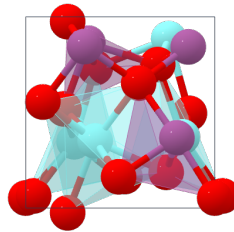
$\text{Na}_2\text{ZnGeO}_4$
 $\text{Li}_2\text{AlSiO}_4$
 -2.85 eV/atom



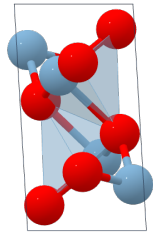
LiSiNO
 $\text{Li}_2\text{Si}_2\text{N}_2\text{O}_2\text{F}$
 -2.21 eV/atom



$\text{CdCu}_2\text{GeS}_4$
 $\text{MgAl}_6\text{O}_{10}$
 -2.87 eV/atom



CeAlO_3
 YScO_3
 -3.75 eV/atom



Ga_2O_3
 Al_2O_3
 -3.29 eV/atom

Figure D.2: Visualization of the final structures obtained by LLMatDesign for the formation energy task. These structures represent the ones with the minimum formation energy per atom from the first run of all 10 starting materials. The chemical formulae in red represent the starting materials, followed by the formulae of the structures with the lowest formation energies and their corresponding formation energy per atom values. GPT-4o with history is utilized as the LLM engine.

E DFT Calculations

Table E.1: DFT results for lowest-energy structures obtained from the formation energy task, averaged across all starting materials and runs.

	GPT-4o with history	Random
Formation energy per atom (eV/atom)	−2.31	−1.51
Job success rate (%)	73.3	40.0